

Aeromechanical Model for Robobee Simulator

Much of what is found here can be found in Whitney and Wood’s 2010 paper, ”Aeromechanics of passive rotation in flapping flight”.

The overall simulation framework looks like

- Grab current wing/body state
- Hinge moment balance
- Aerodynamic force and moment calculation.
- Applied Drake forces/torques
- Integration

1 Methods

1.1 Current Kinematic State

Note that ϕ is the stroke angle (the angle as the wing sweeps back and forth), ψ is the passive wing pitch angle, and θ is the deviation angle (the angle as the wing moves up and down). The total wing angular velocity can be expressed as a vector sum, where

$$\boldsymbol{\omega} = -\dot{\phi}\mathbf{e}_X + \dot{\theta}\mathbf{e}_z + \dot{\psi}\mathbf{e}_x. \quad (1)$$

But since there is no DOF in the θ direction, we can set them to 0. Then, the equation simply means it is a sum of the sweep angular velocity and the pitch angle velocity. We then want to break apart $\boldsymbol{\omega}$ into $\boldsymbol{\omega}_h$ and $\boldsymbol{\omega}_x$ (angular velocity around the wing’s own pitch axis). Then, we can define

$$\boldsymbol{\omega}_h = \boldsymbol{\omega} - \boldsymbol{\omega}_x\mathbf{e}_x. \quad (2)$$

(which is literally just saying removing the $\mathbf{e}_x$ component from rotation). Then, for the no stroke-plane-deviation case,

$$\boldsymbol{\omega}_h = -\dot{\phi}. \quad (3)$$

And after a change of basis (see Appendix 4.1), we can express the angular velocity in the wing-fixed frame as

$$\boldsymbol{\omega}_x = \dot{\psi} - \dot{\phi}\sin\theta \quad (4)$$

$$= \dot{\psi}, \quad (5)$$

again because $\theta = 0$ in our simplified planar case.

1.1.1 Blade-element kinematics and force application

The Drake/CAD/URDF wing body frame B is not necessarily aligned with the aerodynamic span/chord axes used by the blade-element model. Let $\theta_s = \text{span_axis_angle_rad}$ be the fixed angle between the body-frame x -axis and the aerodynamic span axis. Expressed in the wing body frame,

$$\hat{\mathbf{s}}_B = \cos\theta_s \hat{\mathbf{x}}_B + \sin\theta_s \hat{\mathbf{y}}_B, \quad (6)$$

$$\hat{\mathbf{c}}_B = \sigma_c (-\sin\theta_s \hat{\mathbf{x}}_B + \cos\theta_s \hat{\mathbf{y}}_B), \quad (7)$$

$$\hat{\mathbf{n}}_B = \hat{\mathbf{s}}_B \times \hat{\mathbf{c}}_B, \quad (8)$$

where

$$\sigma_c = \text{sign}(\text{chord_axis_sign}).$$

The body-to-aerodynamic-component projection matrix is

$$X_{AB} = \begin{bmatrix} \hat{\mathbf{s}}_B^T \\ \hat{\mathbf{c}}_B^T \\ \hat{\mathbf{n}}_B^T \end{bmatrix}. \quad (9)$$

Thus, for any vector $\mathbf{v}^B$ expressed in the wing body frame,

$$\mathbf{v}^A = X_{AB} \mathbf{v}^B = \begin{bmatrix} \mathbf{v}^B \cdot \hat{\mathbf{s}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{c}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{n}}_B \end{bmatrix}. \quad (10)$$

The matrix X_{AB} is fixed for a given wing geometry. The physical pose of the wing is still given by R_{WB} . Therefore aerodynamic basis vectors are mapped to world coordinates by

$$\hat{\mathbf{s}}_W = R_{WB} \hat{\mathbf{s}}_B, \quad \hat{\mathbf{c}}_W = R_{WB} \hat{\mathbf{c}}_B, \quad \hat{\mathbf{n}}_W = R_{WB} \hat{\mathbf{n}}_B. \quad (11)$$

Equivalently, if

$$R_{BA} = [\hat{\mathbf{s}}_B \quad \hat{\mathbf{c}}_B \quad \hat{\mathbf{n}}_B],$$

then

$$X_{AB} = R_{BA}^T, \quad R_{WA} = R_{WB} X_{AB}^T. \quad (12)$$

For blade element i , let r_i be the radial location of the strip and Δr_i its width. The local dynamic pressure and angle of attack are evaluated at a representative blade point P_i . In this model, P_i is chosen on the pitch/hinge line:

$$\mathbf{p}_{B_o P_i}^A = \begin{bmatrix} r_i + x_r \\ 0 \\ 0 \end{bmatrix}. \quad (13)$$

Mapping this point into the body and world frames gives

$$\mathbf{p}_{B_o P_i}^B = X_{AB}^T \mathbf{p}_{B_o P_i}^A, \quad (14)$$

$$\mathbf{p}_{B_o P_i}^W = R_{WB} \mathbf{p}_{B_o P_i}^B. \quad (15)$$

The blade-point velocity is then

$$\mathbf{v}_{P_i}^W = \mathbf{v}_{B_o}^W + \boldsymbol{\omega}_h^W \times \mathbf{p}_{B_o P_i}^W. \quad (16)$$

Here $\boldsymbol{\omega}_h^W$ is the wing angular velocity with the span-axis component removed:

$$\boldsymbol{\omega}_h^W = \boldsymbol{\omega}_B^W - (\boldsymbol{\omega}_B^W \cdot \hat{\mathbf{s}}_W) \hat{\mathbf{s}}_W. \quad (17)$$

The local relative air velocity is first computed in world coordinates and then expressed in the wing body frame:

$$\mathbf{v}_{\text{rel},i}^W = \mathbf{v}_{P_i}^W - \mathbf{v}_{\text{wind}}^W, \quad (18)$$

$$\mathbf{v}_{\text{rel},i}^B = R_{BW} \mathbf{v}_{\text{rel},i}^W. \quad (19)$$

It is then projected into aerodynamic coordinates using the fixed body-to-aero projection matrix:

$$\mathbf{v}_{\text{rel},i}^A = X_{AB} \mathbf{v}_{\text{rel},i}^B = \begin{bmatrix} v_{s,i} \\ v_{c,i} \\ v_{n,i} \end{bmatrix}. \quad (20)$$

The blade-element model uses only the chord-normal part of this relative velocity, so

$$u_i = v_{c,i}, \quad (21)$$

$$w_i = v_{n,i}, \quad (22)$$

$$V_i^2 = u_i^2 + w_i^2, \quad (23)$$

$$\alpha_i = \text{atan2}(w_i, u_i). \quad (24)$$

Using the quasi-steady coefficient model,

$$C_L(\alpha_i) = C_{L,\max} \sin(2\alpha_i), \quad (25)$$

$$C_D(\alpha_i) = \frac{C_{D,\max} + C_{D,0}}{2} - \frac{C_{D,\max} - C_{D,0}}{2} \cos(2\alpha_i), \quad (26)$$

the strip lift and drag magnitudes are

$$\Delta L_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_L(\alpha_i), \quad (27)$$

$$\Delta D_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_D(\alpha_i). \quad (28)$$

In aerodynamic coordinates, the span, chord, and normal unit vectors are simply

$$\hat{\mathbf{e}}_s^A = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_c^A = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_n^A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The drag direction is opposite the local chord-normal relative velocity:

$$\hat{\mathbf{d}}_i^A = -\frac{u_i \hat{\mathbf{e}}_c^A + w_i \hat{\mathbf{e}}_n^A}{\sqrt{u_i^2 + w_i^2}}. \quad (29)$$

The lift direction is perpendicular to both the span axis and the drag direction:

$$\hat{\mathbf{l}}_i^A = \frac{\hat{\mathbf{e}}_s^A \times \hat{\mathbf{d}}_i^A}{\|\hat{\mathbf{e}}_s^A \times \hat{\mathbf{d}}_i^A\|}. \quad (30)$$

Thus the aerodynamic force on strip i , expressed in aerodynamic coordinates, is

$$\mathbf{F}_{i,\text{aero}}^A = \Delta L_i \hat{\mathbf{l}}_i^A + \Delta D_i \hat{\mathbf{d}}_i^A. \quad (31)$$

The force is applied at the strip center of pressure Q_i , not necessarily at the velocity-evaluation point P_i . The nondimensional distance from the leading edge to the center of pressure is modeled as

$$\hat{d}_{cp,i} = \frac{0.82}{\pi} |\alpha_i| + 0.05. \quad (32)$$

If $q_{le,i}$ is the chord-axis location of the leading edge relative to the wing root convention, then

$$y_{le,i} = \hat{y}_r \bar{c} + q_{le,i}, \quad (33)$$

$$y_{cp,i} = y_{le,i} - c_i \hat{d}_{cp,i}. \quad (34)$$

Thus the force application point is naturally written in aerodynamic coordinates as

$$\mathbf{p}_{B_o Q_i}^A = \begin{bmatrix} r_i + x_r \\ y_{cp,i} \\ 0 \end{bmatrix}. \quad (35)$$

Mapping this point into the body and world frames gives

$$\mathbf{p}_{B_o Q_i}^B = X_{AB}^T \mathbf{p}_{B_o Q_i}^A, \quad (36)$$

$$\mathbf{p}_{B_o Q_i}^W = R_{WB} \mathbf{p}_{B_o Q_i}^B. \quad (37)$$

The strip force is mapped to body and world coordinates by

$$\mathbf{F}_{i,\text{aero}}^B = X_{AB}^T \mathbf{F}_{i,\text{aero}}^A, \quad (38)$$

$$\mathbf{F}_{i,\text{aero}}^W = R_{WB} \mathbf{F}_{i,\text{aero}}^B. \quad (39)$$

Equivalently,

$$\mathbf{F}_{i,\text{aero}}^W = R_{WB} X_{AB}^T \mathbf{F}_{i,\text{aero}}^A = R_{WA} \mathbf{F}_{i,\text{aero}}^A. \quad (40)$$

Finally, the total aerodynamic force and moment about the wing body origin are

$$\mathbf{F}_{\text{blade}}^W = \sum_i \mathbf{F}_{i,\text{aero}}^W, \quad (41)$$

$$\mathbf{M}_{\text{blade}}^W = \sum_i \mathbf{p}_{B_o P_i}^W \times \mathbf{F}_{i,\text{aero}}^W. \quad (42)$$

The scalar aerodynamic pitch moment about the wing span/pitch axis is obtained by projecting this moment onto $\hat{\mathbf{s}}_W$:

$$M_{x,\text{aero}} = \mathbf{M}_{\text{blade}}^W \cdot \hat{\mathbf{s}}_W. \quad (43)$$

1.1.2 Blade Velocity

The Drake/CAD/URDF wing body frame B is not necessarily aligned with the aerodynamic span/chord axes used by the blade-element model. Let $\theta_s = \text{span_axis_angle_rad}$ be the fixed angle between the body-frame x -axis and the aerodynamic span axis. Then, expressed in the wing body frame,

$$\hat{\mathbf{s}}_B = \cos \theta_s \hat{\mathbf{x}}_B + \sin \theta_s \hat{\mathbf{y}}_B, \quad (44)$$

$$\hat{\mathbf{c}}_B = \sigma_c (-\sin \theta_s \hat{\mathbf{x}}_B + \cos \theta_s \hat{\mathbf{y}}_B), \quad (45)$$

$$\hat{\mathbf{n}}_B = \hat{\mathbf{s}}_B \times \hat{\mathbf{c}}_B, \quad (46)$$

where $\sigma_c = \text{sign}(\text{chord_axis_sign})$. The body-to-aerodynamic-component projection matrix is

$$X_{AB} = \begin{bmatrix} \hat{\mathbf{s}}_B^T \\ \hat{\mathbf{c}}_B^T \\ \hat{\mathbf{n}}_B^T \end{bmatrix}. \quad (47)$$

Thus, for any vector $\mathbf{v}^B$ expressed in the wing body frame,

$$\mathbf{v}^A = X_{AB} \mathbf{v}^B = \begin{bmatrix} \mathbf{v}^B \cdot \hat{\mathbf{s}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{c}}_B \\ \mathbf{v}^B \cdot \hat{\mathbf{n}}_B \end{bmatrix}. \quad (48)$$

The world rotation R_{WB} is still the physical pose of the wing body frame. The span-axis angle only changes the aerodynamic basis vectors inside that body frame, so aerodynamic vectors are mapped to the world frame by first expressing them in body coordinates:

$$\mathbf{v}^W = R_{WB} \mathbf{v}^B. \quad (49)$$

For example,

$$\hat{\mathbf{s}}_W = R_{WB} \hat{\mathbf{s}}_B, \quad \hat{\mathbf{c}}_W = R_{WB} \hat{\mathbf{c}}_B, \quad \hat{\mathbf{n}}_W = R_{WB} \hat{\mathbf{n}}_B. \quad (50)$$

Equivalently, if R_{BA} has the aerodynamic basis vectors as columns, then

$$R_{BA} = [\hat{\mathbf{s}}_B \quad \hat{\mathbf{c}}_B \quad \hat{\mathbf{n}}_B], \quad (51)$$

$$X_{AB} = R_{BA}^T, \quad (52)$$

$$R_{WA} = R_{WB} X_{AB}^T. \quad (53)$$

For wing strip i , define the blade point P_i in the wing body frame as

$$\mathbf{p}_{B_o P_i}^B = (r_i + x_r) \hat{\mathbf{s}}_B + y_{h,i} \bar{\mathbf{c}} \hat{\mathbf{c}}_B. \quad (54)$$

The corresponding world-frame offset and blade-point velocity are

$$\mathbf{p}_{B_o P_i}^W = R_{WB} \mathbf{p}_{B_o P_i}^B, \quad (55)$$

$$\mathbf{v}_{P_i}^W = \mathbf{v}_{B_o}^W + \boldsymbol{\omega}_h^W \times \mathbf{p}_{B_o P_i}^W. \quad (56)$$

As above, $\boldsymbol{\omega}_h^W$ is the angular velocity with the span-axis component removed:

$$\boldsymbol{\omega}_h^W = \boldsymbol{\omega}_B^W - (\boldsymbol{\omega}_B^W \cdot \hat{\mathbf{s}}_W) \hat{\mathbf{s}}_W. \quad (57)$$

The relative air velocity is then transformed back into the wing body frame:

$$\mathbf{v}_{\text{rel},i}^W = \mathbf{v}_{P_i}^W - \mathbf{v}_{\text{wind}}^W, \quad (58)$$

$$\mathbf{v}_{\text{rel},i}^B = R_{BW} \mathbf{v}_{\text{rel},i}^W. \quad (59)$$

Projecting onto the aerodynamic chord-normal plane gives

$$u_i = \mathbf{v}_{\text{rel},i}^B \cdot \hat{\mathbf{c}}_B, \quad (60)$$

$$w_i = \mathbf{v}_{\text{rel},i}^B \cdot \hat{\mathbf{n}}_B, \quad (61)$$

$$V_i^2 = u_i^2 + w_i^2, \quad (62)$$

$$\alpha_i = \text{atan2}(w_i, u_i). \quad (63)$$

Using the quasi-steady coefficient model,

$$C_L(\alpha_i) = C_{L,\text{max}} \sin(2\alpha_i), \quad (64)$$

$$C_D(\alpha_i) = \frac{C_{D,\text{max}} + C_{D,0}}{2} - \frac{C_{D,\text{max}} - C_{D,0}}{2} \cos(2\alpha_i), \quad (65)$$

so the strip lift and drag magnitudes are

$$dL_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_L(\alpha_i), \quad (66)$$

$$dD_i = \frac{1}{2} \rho V_i^2 c_i \Delta r_i C_D(\alpha_i). \quad (67)$$

The drag direction is opposite the local chord-normal relative velocity, and the lift direction is perpendicular to both the span axis and the drag direction:

$$\hat{\mathbf{d}}_i^B = -\frac{u_i \hat{\mathbf{c}}_B + w_i \hat{\mathbf{n}}_B}{\sqrt{u_i^2 + w_i^2}}, \quad (68)$$

$$\hat{\mathbf{l}}_i^B = \frac{\hat{\mathbf{s}}_B \times \hat{\mathbf{d}}_i^B}{\|\hat{\mathbf{s}}_B \times \hat{\mathbf{d}}_i^B\|}. \quad (69)$$

Therefore, the aerodynamic force on strip i , expressed in the wing body frame, is

$$\mathbf{F}_{i,\text{aero}}^B = dL_i \hat{\mathbf{l}}_i^B + dD_i \hat{\mathbf{d}}_i^B. \quad (70)$$

The force application point is the center of pressure. First compute

$$\hat{d}_{cp,i} = \frac{0.82}{\pi} |\alpha_i| + 0.05, \quad (71)$$

$$y_{le,i} = \hat{y}_r \bar{c} + q_{le,i}, \quad (72)$$

$$y_{cp,i} = y_{le,i} - c_i \hat{d}_{cp,i}. \quad (73)$$

Then the point where the strip force is applied is

$$\mathbf{p}_{B_o Q_i}^B = (r_i + x_r) \hat{\mathbf{s}}_B + y_{cp,i} \hat{\mathbf{c}}_B. \quad (74)$$

Finally, the strip force is mapped to world coordinates before summing over blade elements:

$$\mathbf{F}_{i,\text{aero}}^W = R_{WB} \mathbf{F}_{i,\text{aero}}^B, \quad (75)$$

$$\mathbf{F}_{\text{blade}}^W = \sum_i \mathbf{F}_{i,\text{aero}}^W. \quad (76)$$

1.2 Compute Aerodynamic Coefficients

1.2.1 Computing α

$$\alpha = f(\psi, \dot{\phi}, \text{stroke direction}, \text{body velocity}, \text{wind}) \quad (77)$$

If W is the world frame,

$$\mathbf{v}_{rel}^W = \mathbf{v}_{wing}^W - \mathbf{v}_{air}^W \quad (78)$$

$$\mathbf{v}_{rel}^{wing} = R_W^{wing} \mathbf{v}_{rel}^W \quad (79)$$

$$\alpha = \text{atan2}(v_{normal}, v_{chord}) \quad (80)$$

But for wing velocity **through** air, then

$$\alpha = \text{atan2}(v_{normal}, v_{chord}) \quad (81)$$

Another quick note:

$$O' = O + (x_r, y_r). \quad (82)$$

See Fig. 2 for the coordinate axes superimposed on the CAD model. For the latest CAD model, that equals 3.73, -0.86 mm.

1.2.2 Computing Aerodynamic Coefficients

We make the quasi-steady assumption, where we pretend that in every instant, the wing is in a steady flow with its current velocity and angle of attack. An empirical fit for the coefficient of lift gives us

$$C_L(\alpha) = C_{L,\max} \sin(2\alpha). \quad (83)$$

For the coefficient of drag, we have

$$C_D(\alpha) = \frac{C_{D,\max} + C_{D,0}}{2} - \frac{C_{D,\max} - C_{D,0}}{2} \cos(2\alpha). \quad (84)$$

For *Drosophila*, we have $C_{L,\max} = 1.8$, $C_{D,\max} = 3.4$, and $C_{D,0} = 0.4$, from Dickinson et al. (1999).

The normal-force coefficient, which describes the coefficient for force normal to wing surface (and ultimately generates a moment about the pitch hinge), is given by

$$C_N = C_L \cos(\alpha) + C_D \sin(\alpha). \quad (85)$$

1.3 Compute blade-element lift and drag

Before starting, note the sign conventions, where $+\phi$, $+\omega_x$, and $+\alpha$ are all positive for the local $+x$ axis of the wing (i.e. pointing away from the root, along the hinge). For each strip, shown in Fig. 1, we have

$$dF_L = \frac{1}{2} \rho V_i^2 C_L(\alpha) c(r') dr, \quad (86)$$

$$dF_D = \frac{1}{2} \rho V_i^2 C_D(\alpha) c(r') dr, \quad (87)$$

where $V_i = |\omega_h|(r + x_r)$. To derive this velocity, observe that a blade element has radial distance $r + x_r$ (see Fig. 1). Then, the tangential speed (around the stroke, X' axis), is $u(r) = |\omega_h|(r + x_r)$, so $V_i^2 = \omega_h^2 (r + x_r)^2$. Furthermore, $c(r')$ is the chord length at the blade element, where r' is the distance from the wing root

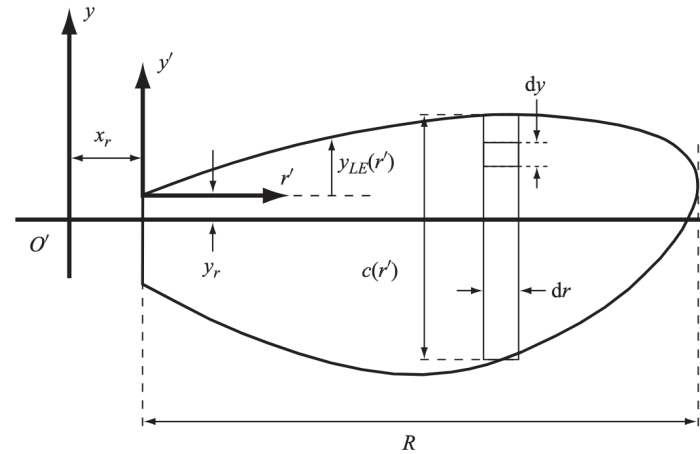


Figure 1: Aeromechanical model of the RoboBee.

to the blade element. Then, the total lift is the integral over the wing span:

$$F_L = \int_0^R dF_L \quad (88)$$

$$= \frac{1}{2} \rho \omega_h^2 C_L(\alpha) \underbrace{\int_0^R (r + x_r)^2 c(r') dr}_{\hat{F}}, \quad (89)$$

where $\hat{F}$ is the dimensionless wing-shape factor. Full derivation can be found in Appendix 4.2. To non-dimensionalize the integral, set

$$\hat{r} = \frac{r'}{R}, \quad (90)$$

$$\hat{x}_r = \frac{x_r}{R}, \quad (91)$$

$$\hat{c}(\hat{r}) = \frac{c(r')}{\bar{c}}, \quad (92)$$

where $\bar{c}$ is the mean chord length $\frac{A}{R}$. (R is the wing span).

1.4 Compute aerodynamic pitch moment about the hinge

$M_{x,\text{aero}}$ is combination of lift and drag forces, but as a moment. While the integral form given by Whitney and Wood is

$$M_{x,\text{aero}} = \frac{1}{2} \rho \omega_h^2 C_N(\alpha) \int_0^R y_{cp}(r') (r' + x_r)^2 c(r') dr'. \quad (93)$$

Note that α , the angle of attack. In discrete form, we have

$$M_{x,\text{aero}} \approx \sum_{i=1}^N y_{cp,i} dF_{N,i}. \quad (94)$$

Substituting in Eq. 86, then we have

$$M_{x,\text{aero}} \approx - \sum_{i=1}^N \text{sgn}(\alpha) y_{cp,i} \frac{1}{2} \rho V_i^2 C_N(\alpha_i) c_i \Delta r_i. \quad (95)$$

Note the inclusion of sgn because $\mathbf{F}_N = F_N \mathbf{e}_N = -\text{sgn}(\alpha) F_N \mathbf{e}_z$.

Here, $y_{cp}(r')$ is the distance from the leading edge to the center of pressure for the blade element at r' , and is given by

$$y_{cp} = y_r + y_{LE}(r') - c(r') \hat{d}_{cp}. \quad (96)$$

y_{LE} is basically the distance from the leading edge to the rotation axis, while $\hat{d}_{cp}$ is the dimensionless distance (think: fraction) from the leading edge to the center of pressure, which is given by Dickson et al. (2006):

$$\hat{d}_{cp} = \frac{0.82}{\pi} |\alpha| + 0.05. \quad (97)$$

1.5 Compute passive hinge restoring moment

$M_{x,\text{hinge}} = -\kappa_H \psi$ is a simple linear torsional spring model, where k is the torsional stiffness. The torsional stiffness is given by

$$\kappa_H = \frac{E_h t_h^3 w_h}{12 L_h}, \quad (98)$$

where $t_h = 0.0125$ mm, $w_h = 2.7$ mm, and $L_h = 0.10$ mm are the thickness, width, and length of the central layer. For Kapton, $E_h = 2.5$ GPa.

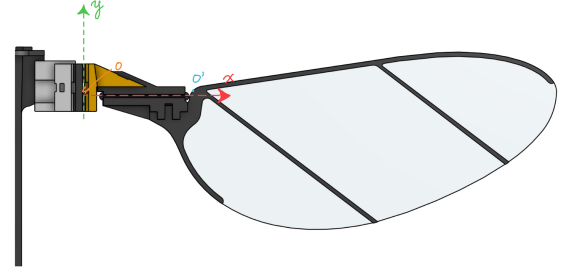


Figure 2: Coordinate axes superimposed on CAD model.

1.6 Compute aerodynamic damping moment

$$M_{x,\text{rd}} = -\frac{1}{2}\rho\omega_x|\omega_x|C_{\text{rd}}\bar{c}^4 R\hat{Y}_{\text{rd}}. \quad (99)$$

where

$$\hat{Y}_{\text{rd}} = \int_0^1 \hat{y}_{\text{rd}}(\hat{r}) d\hat{r}. \quad (100)$$

and

$$\hat{y}_{\text{rd}}(\hat{r}) = \frac{1}{4} \left[|\hat{y}_r + \hat{y}_{LE}| (\hat{y}_r + \hat{y}_{LE})^3 - |\hat{y}_r + \hat{y}_{LE} - \hat{c}| (\hat{y}_r + \hat{y}_{LE} - \hat{c})^3 \right], \quad (101)$$

the non-dimensional location of the effective moment arm of the rotational damping term. Set $C_{\text{rd}} = 2.0$ as the rotational damping moment coefficient. Again, $\hat{y}_r$ is the dimensionless distance from the rotation axis to the root of the wing, while $\hat{y}_{LE}$ is the dimensionless distance from the leading edge (outline of the wing planiform) to the rotation axis.

1.7 Compute added mass effect moment

The air inertia resisting wing acceleration, i.e. aerodynamic forces and moments that depend on the body's acceleration, adds another term to the total moment. Whitney and Wood start from the two-dimensional added-mass force and moment on a thin section. After dropping cross-terms that duplicate other quasi-steady blade-element terms, the reduced section equations are

$$Z_0 = -\lambda_z \dot{W}_0 - \lambda_{z\omega} \dot{\omega}_x, \quad (102)$$

$$Y_0 = 0, \quad (103)$$

$$M_0 = -\lambda_{z\omega} \dot{W}_0 - \lambda_\omega \dot{\omega}_x, \quad (104)$$

where Z_0 is the added-mass normal force per unit span, M_0 is the added-mass pitch moment per unit span, and

$$\lambda_z = \pi \rho a^2, \quad (105)$$

$$\lambda_{z\omega} = -\pi \rho a^2 y_h, \quad (106)$$

$$\lambda_\omega = \pi \rho a^2 y_h^2 + \frac{1}{8} \pi \rho a^4. \quad (107)$$

Here $a = c(r')/2$ is the local semi-chord, and y_h is the signed offset from the torsional axis to the mid-chord of the local blade-element strip.

Using Whitney and Wood's kinematics, the local normal acceleration of the hinge line is

$$\dot{W}_0(r') = -(r' + x_r) (\dot{\omega}_y - \omega_x \omega_z), \quad (108)$$

where $(r' + x_r)$ is the distance from the stroke-axis shoulder to the blade element. Radial integration of the reduced section moment gives

$$M_{x,\text{am}} = -\left(\frac{\pi}{4} \rho \bar{c}^3 R^2 \hat{I}_{xy,\text{am}}\right) (\dot{\omega}_y - \omega_x \omega_z) - \left(\frac{\pi}{4} \rho \bar{c}^4 R \hat{I}_{xx,\text{am}}\right) \dot{\omega}_x. \quad (109)$$

Here, $\hat{I}_{xy,\text{am}}$ and $\hat{I}_{xx,\text{am}}$ are added-mass geometry integrals analogous to inertial products, and are constants found by

$$\hat{I}_{xy,\text{am}} = \int_0^1 (\hat{r} + \hat{x}_r) \hat{c}(\hat{r})^2 \hat{y}_h(\hat{r}) d\hat{r}, \quad (110)$$

$$\hat{I}_{xx,\text{am}} = \int_0^1 \hat{c}(\hat{r})^2 \left(\hat{y}_h(\hat{r})^2 + \frac{1}{32} \hat{c}(\hat{r})^2 \right) d\hat{r}. \quad (111)$$

1.7.1 Applying added mass as a blade-element normal force

For a simulator that applies forces directly to the wing, the translational part of the added-mass effect can be applied strip-by-strip instead of only as a pre-integrated pitch moment. For blade element i , with radial width Δr_i , local

chord c_i , semi-chord $a_i = c_i/2$, and signed mid-chord offset $y_{h,i}$, compute

$$\lambda_{z,i} = \pi \rho a_i^2, \quad (112)$$

$$\lambda_{z\omega,i} = -\pi \rho a_i^2 y_{h,i}, \quad (113)$$

$$\dot{W}_{0,i} = -(r'_i + x_r)(\dot{\omega}_y - \omega_x \omega_z), \quad (114)$$

$$Z_{0,i} = -\lambda_{z,i} \dot{W}_{0,i} - \lambda_{z\omega,i} \dot{\omega}_x. \quad (115)$$

The normal added-mass force on that strip is then

$$\Delta \mathbf{F}_{am,i} = Z_{0,i} \Delta r_i \mathbf{e}_z, \quad (116)$$

with the sign of $Z_{0,i}$ carrying the direction along the local wing-fixed normal $\mathbf{e}_z$. Depending on the simulator's force convention, this may need to be negated when converting from “force of the air on the wing” to “reaction force of the wing on the body.”

The natural application point for this strip force is the local mid-chord location used by the added-mass section model,

$$\mathbf{p}_{am,i} = (r'_i + x_r) \mathbf{e}_x + y_{h,i} \mathbf{e}_y, \quad (117)$$

measured from the shoulder/hinge-frame origin. Applying Eq. 116 at Eq. 117 gives a pitch-axis moment contribution

$$\Delta M_{x,am,i}^{(F)} = y_{h,i} Z_{0,i} \Delta r_i. \quad (118)$$

This captures the moment induced by the translational added-mass normal force. To reproduce the full Whitney–Wood added-mass pitch moment, also include the local pure rotational added-mass couple

$$\Delta M_{x,am,i}^{(\omega)} = -\lambda_{\omega,i} \dot{\omega}_x \Delta r_i, \quad (119)$$

where

$$\lambda_{\omega,i} = \pi \rho a_i^2 y_{h,i}^2 + \frac{1}{8} \pi \rho a_i^4. \quad (120)$$

Thus, if using point forces only, Eq. 116 is the translational/normal added-mass load to apply to the strip. If matching the analytical rotational dynamics exactly, apply both the strip force and the additional strip torque/couple $\Delta M_{x,am,i}^{(\omega)}$. Do not also add the pre-integrated $M_{x,am}$ term as a scalar pitch moment when the strip force is active, or the force-induced moment will be double counted.

1.8 Integration

$$I_{xx} \ddot{\psi} = M_x + I_{xy} \ddot{\phi} \cos \psi + \frac{1}{2} I_{xx} \dot{\phi}^2 \sin(2\psi). \quad (121)$$

$$\ddot{\psi} = \frac{M_x + I_{xy} \ddot{\phi} \cos \psi + \frac{1}{2} I_{xx} \dot{\phi}^2 \sin(2\psi)}{I_{xx}}, \quad (122)$$

where $\ddot{\phi}$ is the stroke angular acceleration, which we can compute from the current stroke angle and velocity/actuator input. See Appendix 5 for derivation of I_{xx} and I_{xy} . In particular, assume $\phi(t)$ (taken at transmission link 1) is commanded from the control algorithms (typically a sin wave with $\omega = 170\text{Hz}$, i.e at resonance of transmission mechanism). This should be parametrized in the code. If added mass is applied as blade-element normal forces, the remaining scalar total moment is

$$M_{\text{total}} = M_{\text{aero}} + M_{\text{rot}} + M_{\text{hinge}} \quad (123)$$

But since aero forces originate as translational forces applied to the center of pressure, we can't just apply them purely as moments.

So really, M_{aero} needs to be applied as

$$\sum_i \mathbf{r}_{cp,i} \times d\mathbf{F}_i, \quad (124)$$

where

$$\mathbf{r}_{cp,i} = (r + x_r) \mathbf{e}_x + y_{cp}(r) \mathbf{e}_y. \quad (125)$$

2 Lumped-Parameter Actuation Dynamics and Stroke Frequency Response

Everything above treats the stroke angle $\phi(t)$ as given. In the original code it effectively was: the wing voltage mapped algebraically to a slider displacement ($2.5 \mu\text{m/V}$ about a 100 V rest point) and a stiff PD joint actuator forced the slider to track it, so the stroke was kinematically prescribed and a frequency sweep would have measured the tracking bandwidth of that servo rather than any physics. Both builds now use the lumped force-driven actuator described below; this section is kept as the record of what was added and why. This section records how the transmission and actuator are modeled as a lumped second-order system in the literature, which of those lumped terms Drake already reproduces implicitly, and what has to be added so that the simulator can produce a stroke-amplitude frequency response $|\Phi/V_{pp}|(\omega)$ of the kind shown in Steinmeyer et al., “Yaw Torque Authority for a Flapping-Wing Micro-Aerial Vehicle” (ICRA 2019), Fig. 3.

2.1 Reference lumped model

Following Finio et al. (2011) and Jafferis et al. (2016), Steinmeyer et al. model the actuator, transmission, and wing as one spring–mass–damper written on the actuator (slider) side. With actuator effective mass m_a , actuator stiffness k_a , transmission ratio T (rad of stroke per metre of slider travel), transmission flexure stiffness k_t , wing moment of inertia about the stroke axis J_ϕ , and a linearized aerodynamic damping coefficient b acting at radius r_{cp} ,

$$m_{eq} = m_a + T^2 J_\phi, \quad (126)$$

$$b_{eq} = T^2 r_{cp} b, \quad (127)$$

$$k_{eq} = k_a + T^2 k_t, \quad (128)$$

and the voltage-to-stroke transfer function is

$$H(\omega) = \frac{\Phi(\omega)}{V(\omega)} = \frac{A_H}{m_{eq}(\omega i)^2 + b_{eq}(\omega i) + k_{eq}}, \quad (129)$$

where A_H converts volts to slider force. Their Table I values are $m_a = 25 \text{ mg}$, $J_\phi = 51.1 \text{ mg mm}^2$, $T = 2666$, $r_{cp} = 1.42 \times 9.56 \text{ mm}$, $b = 2.03 \times 10^{-6} \mu\text{Ns/m}$, $k_a = 300 \text{ N/m}$, $k_t = 28.2 \mu\text{N m/rad}$, giving $m_{eq} = 0.388 \text{ g}$, $b_{eq} = 0.196 \text{ N s/m}$, $k_{eq} = 500.4 \text{ N/m}$ and a resonance near $f_n = \frac{1}{2\pi} \sqrt{k_{eq}/m_{eq}} \approx 181 \text{ Hz}$. Note that the wing term dominates: $T^2 J_\phi \approx 363 \text{ mg}$ versus $m_a = 25 \text{ mg}$, so about 94% of the equivalent mass is wing inertia reflected through the transmission. The resonant frequency is set almost entirely by J_ϕ against $T^2 k_t$.

This exact model is already implemented in MATLAB as `simulink/system id/get_transfer_function.m` with the Table I parameters in `data_optimize_full_optimization.m`, but it is only ever evaluated at two discrete frequencies and is not connected to the Drake plant. Plotting Eq. (129) over 50–400 Hz from that function is the analytic reference curve that the simulator result below should be compared against.

2.2 Mapping of lumped terms onto the Drake plant

The point of doing this in Drake rather than with Eq. (129) is that several of the lumped parameters stop being parameters and become consequences of the geometry and the aerodynamics already in the model. Table 1 summarizes the mapping.

Lumped term	Drake representation	Status
T (transmission ratio)	Emergent from four-bar linkage kinematics	Done
J_ϕ (wing inertia)	Rigid-body inertia of hinge/frame/membrane bodies	Done
b (aero damping)	Blade-element drag reacting through linkage	Done
m_a (actuator mass)	Reflected inertia on slider actuator	Done (standalone app and Simulink server)
k_a (actuator stiffness)	<code>PrismaticSpring</code> on slider joint	Done (standalone app and Simulink server)
$A_H V$ (voltage force)	Pure force on slider actuation input	Done (standalone app and Simulink server)
k_t (flexure stiffness)	<code>RevoluteSpring</code> on each of 6 flexure joints	Done (standalone app and Simulink server)

Table 1: Correspondence between the Steinmeyer et al. lumped parameters and the Drake plant.

2.2.1 Transmission ratio (done)

The transmission is a real closed-loop mechanism in the `MultibodyPlant`: `slider_2` (prismatic, the actuator tip surrogate) drives `transmission_link_1`, which connects through `transmission_link_2` and the loop-closure ball constraints to `transmission_hinge`, whose joint `revolute_1_3` is the stroke angle ϕ for the left wing (`slider_1` / `revolute_1_4` for the right). T is therefore the local slope $\partial\phi/\partial x_{slider}$ of the forward kinematics, including its nonlinearity at large stroke, and does not need to be specified.

2.2.2 Wing inertia (done)

J_ϕ enters the lumped model only through m_{eq} . In Drake the same quantity arises because the transmission hinge, hinge base, hinge wing, carbon-fibre frame, and membrane are rigid bodies carried by the stroke joint. Walking the URDF chain from `transmission_hinge` at zero joint angle and applying the parallel-axis theorem about the stroke axis gives, per wing,

Body	mass (mg)	$r_\perp$ from stroke axis (mm)	J (mg mm ²)
wing_cf	0.614	7.42	43.2
wing_membrane	0.077	9.23	7.4
hinge_left_wing	0.514	2.36	3.2
hinge_left_base	0.418	1.76	1.6
transmission_hinge + link_2	1.256	~0.6	0.7
Total	2.879		56.0

Table 2: URDF wing inertia about the stroke axis, left wing; the right wing is identical to three figures. Compare $J_\phi = 51.1$ mg mm² in Steinmeyer et al.

The URDF value is 10% above the paper’s, which by itself would move the predicted resonance from 181 to about 173 Hz. This is within the robot-to-robot spread reported in that paper. Because J_ϕ is 94% of m_{eq} , this table is the single most important input to the resonant frequency and should be re-checked whenever the wing URDF changes. Note also that the added-mass normal force of Section 1.7.1 acts as an additional effective inertia that the lumped model does not separate out; the simulated m_{eq} will read slightly higher than the rigid-body number.

2.2.3 Aerodynamic damping (done, but quadratic)

The blade-element wrench is applied as a full six-component spatial force on the wing body, so its stroke-axis moment reacts back through the linkage to the slider and is exactly the load the actuator must overcome. This is the physical origin of b . Two caveats. First, the only explicitly authored rotational damping term $M_{x,rd}$ acts about the span (pitch) axis, not the stroke axis; stroke damping comes purely from the drag moments $\sum_i \mathbf{r}_{cp,i} \times d\mathbf{F}_i$. Second, drag scales with $\dot{\phi}|\dot{\phi}|$ whereas the lumped b is a linearization. The simulated frequency response is therefore amplitude-dependent: the resonance peak will be taller and narrower at small V_{pp} and flatter at flight voltage. The comparison with Eq. (129) holds at one amplitude only, and the sweep should be run at several.

2.2.4 Actuator mass (to do)

The piezo bimorph layers in the URDF are welded to the airframe and never move; the slider link that stands in for the actuator tip weighs 0.3 mg, not the 25 mg that Steinmeyer et al. use for m_a . Their Table I labels this simply “actuator mass”. The URDF gives each actuator two PZT layers of 10.4 mg plus an unmodeled carbon-fibre core, so 25 mg is evidently the physical mass of the bimorph rather than a tip-referred effective mass. For a uniform cantilever the tip-referred value is $33/140 \approx 0.24$ of the physical mass, i.e. roughly 6 mg; the tapered RoboBee actuator with its rigid tip extension will differ from that fraction but remain well below one. Using the physical mass therefore overestimates m_a by three to four times, which moves f_n by only about 2.5% because m_a is a small share of m_{eq} . Use 25 mg when reproducing the paper’s curve and treat ~ 6 mg as the physically defensible value. The cleanest fix is Drake’s actuator reflected inertia: `JointActuator::set_default_rotor_inertia` and `set_default_gear_ratio` add $I_{rotor} n^2$ to the joint degree of freedom, which for a prismatic joint has units of mass. Setting $I_{rotor} = 25 \times 10^{-6}$ kg and $n = 1$ on each slider actuator adds exactly m_a without altering gravity loads, centre of mass, or the loop-closure geometry. The alternative of raising the `transmission_link_1` mass in the URDF also works but shifts the body COM. Since m_a is only 6% of m_{eq} , omitting it costs about 3% in f_n ; it is included for completeness rather than because it moves the answer.

2.2.5 Actuator stiffness and voltage force (to do)

The standard lumped description of a piezo bending actuator is a spring to ground whose equilibrium shifts with voltage,

$$F_{tip} = k_a(\delta_{free}(V) - x), \quad \delta_{free}(V) = c_V(V - V_{rest}), \quad (130)$$

where x is the actual slider displacement from its rest position (the loaded actuator tip deflection), δ_{free} is the unloaded tip deflection the voltage would produce, and $c_V = 2.5 \times 10^{-6}$ m/V, $V_{rest} = 100$ V as currently coded. So, a spring in parallel with a force source. At $x = 0$ the actuator exerts its blocking force $k_a \delta_{free}$; at $x = \delta_{free}$ it exerts none. Under load the slider settles where F_{tip} balances the reflected transmission and aerodynamic forces, which is why $x_{ss}/\delta_{free} = k_a/k_{eq}$ in Section 2.5. This is not a series arrangement; it decomposes into two elements acting in parallel on the same slider degree of freedom:

- a **PrismaticSpring** force element on each slider joint with stiffness $k_a = 300$ N/m and nominal position equal to the slider rest position read from the default context after **Finalize**;
- a pure force input $u = k_a c_V(V - V_{rest})$ fed to the plant’s actuation input port.

Concretely, the existing slider **JointActuators** are kept but **set_controller_gains** is no longer called, so they become pure force actuators, and the effort limit is set to the blocking force $k_a c_V \Delta V_{max} \approx 300 \times 0.25 \times 10^{-3} \approx 75$ mN instead of 10^9 N. **VoltageStrokeSource** changes from emitting a four-vector desired state $[q_1, q_2, v_1, v_2]$ to emitting a two-vector of forces, and is connected to **get_actuation_input_port()** instead of the desired-state port. The voltage-derivative inputs it currently consumes become unnecessary. The commented-out 0.1 N effort limit already present in the source suggests this direction was anticipated.

2.2.6 Transmission flexure stiffness (done, distributed)

The lumped k_t is one torsional stiffness at the output angle; physically the four-bar has a flexure at every joint. Rather than a single lumped spring, each flexure is modeled individually so that k_t becomes emergent in the same way T already is. Each transmission flexure is a Kapton layer with the same stiffness formula as the wing hinge,

$$\kappa_i = \frac{Et^3w}{12L_i}, \quad E = 2.5 \text{ GPa}, \quad t = 25 \mu\text{m}, \quad w = 0.6 \text{ mm}, \quad (131)$$

with three flexures per side. In the URDF these are real revolute joints, so each carries a **RevoluteSpring** anchored at the joint’s default (CAD rest) angle:

Flexure	Joint (left / right)	L (mm)	κ ($\mu\text{N m/rad}$)
base $\rightarrow$ hinge (stroke output ϕ)	revolute_1.3 / revolute_1.4	0.20	9.77
hinge $\rightarrow$ link_2	revolute_2 / revolute_2.1	0.20	9.77
link_1 $\rightarrow$ link_2	revolute_3.loop_closure / _1	0.14	13.95

Table 3: Transmission flexure springs added by **AddTransmissionFlexureSprings**.

The link_1 $\rightarrow$ link_2 pivot is the one the URDF opened into a loop-closure helper body; that helper is welded back onto link_2 by constraint, so a spring on that joint acts between link_1 and link_2 through the constraint. The effective output stiffness is $k_t^{\text{eff}} = \sum_i \kappa_i (\partial \theta_i / \partial \phi)^2$, which depends on the linkage geometry; the naive sum of the three is $33.5 \mu\text{N m/rad}$, the same order as the paper’s lumped 28.2. The static validation of Section 2.5 measures k_t^{eff} directly; it came out at only 11–17 $\mu\text{N m/rad}$, about half the paper’s lumped value. Since 2026-09-16 a lumped trim spring k_t^{trim} (**kTransmissionOutputStiffnessTrimNmPerRad**, currently 40 $\mu\text{N m/rad}$) is therefore added on each stroke output joint (**revolute_1.3** / **revolute_1.4**) in parallel with the geometric flexure springs, in both builds; setting it to zero recovers the purely geometric model. Its size was chosen from the sweep so that the stroke-amplitude peak at the 131 V_{pp} hover drive sits at 155 Hz, the resonance of the current hardware (Section 2.4), not from flexure geometry.

No damping is authored on the transmission or slider. The real actuator has some material damping; a small slider joint damping may also be needed for numerical reasons once the PD gains that currently stiffen the loop-closure constraints are removed (see Section 2.5).

2.3 Sweep infrastructure (done 2026-09-16)

- The standalone app takes `--drive_frequency=HZ`, `--duration=S`, `--realtime_rate=X` (0 = as fast as possible) and `--slider_log=PATH`. The plant time step is still derived from the drive frequency as $1/(100f_{drive})$, so every swept point has 100 steps per cycle but a different discretization; decoupling the step is left as a follow-up.
- The slider CSV (both builds) now carries `right_stroke_angle_rad` and `left_stroke_angle_rad`: the angles of `revolute_1.4` and `revolute_1.3` relative to their CAD rest angle. The two joints have mirrored positive directions in the URDF, and the right one runs opposite to positive slider displacement.
- `tools/stroke_frequency_sweep.py` runs the welded standalone binary once per (V_{pp}, f) , discards 30 cycles, least-squares fits $c + a \sin \omega t + b \cos \omega t$ to the stroke angle, slider displacement and voltage over the last 10 cycles, and plots gain $|\Phi|/|V|$ (deg/V) and phase against frequency. `--from-csv` redraws the plot from a saved sweep without re-simulating. Each run takes about 0.7 s; 99 runs with `--jobs 4` take about 75 s.

2.4 Measured frequency response

Fig. 3 shows the result over 40–400 Hz in 5 Hz steps at 50, 100, 131 and 200 V_{pp} (actuator side); the data are in `README/stroke_sweep_20260916.csv`.

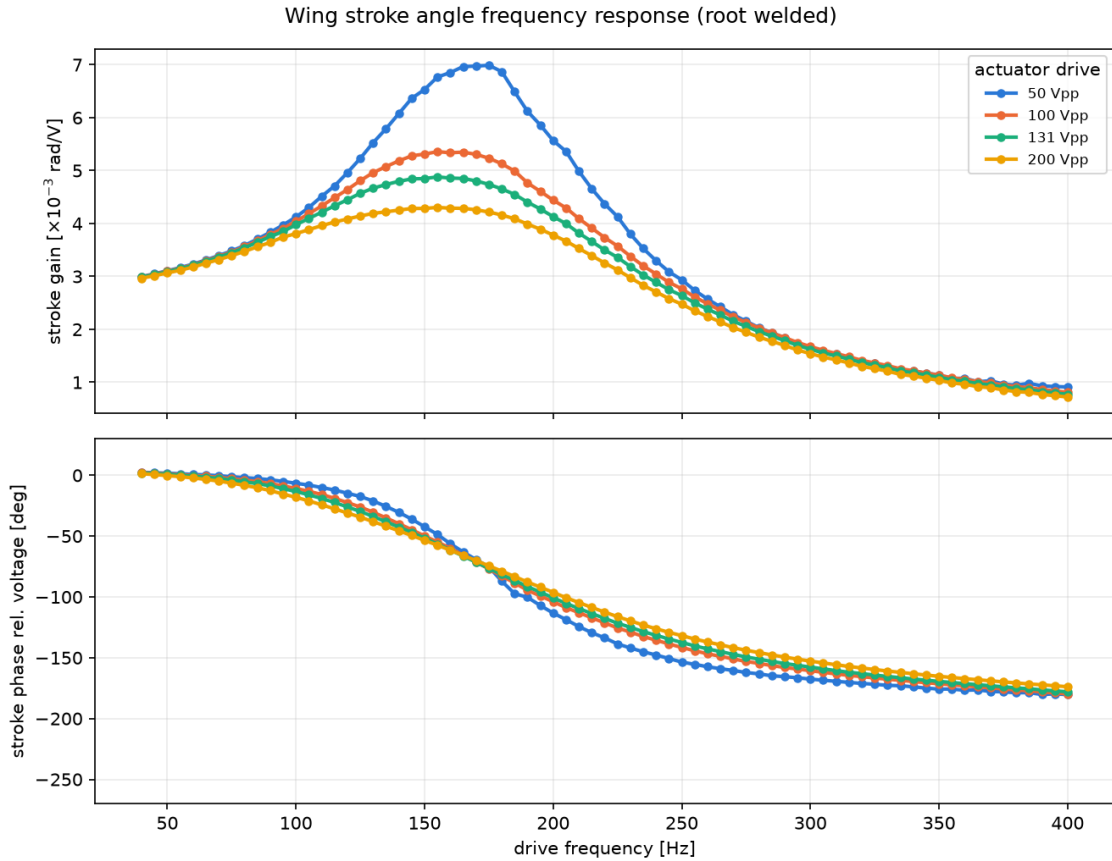


Figure 3: Simulated stroke-angle frequency response with the root welded, at four actuator drive amplitudes; the gain axis is in rad/V with a 10^{-3} multiplier. The Table I lumped model is compared numerically in the text rather than drawn.

We should be careful with figuring out which resonant peak we want to define as the operating frequency.

TODO

In the plots, we can see resonance (ω_r) occur at around 155 Hz for 131 V_{pp} (roughly the voltage for hovering), while the natural frequency (ω_d , the -90° phase crossing) is located around 188 Hz or so (these are the values with the current $40 \mu\text{N m/rad}$ trim; with the earlier $30 \mu\text{N m/rad}$ trim they were 145 and 180 Hz). Recall that resonant

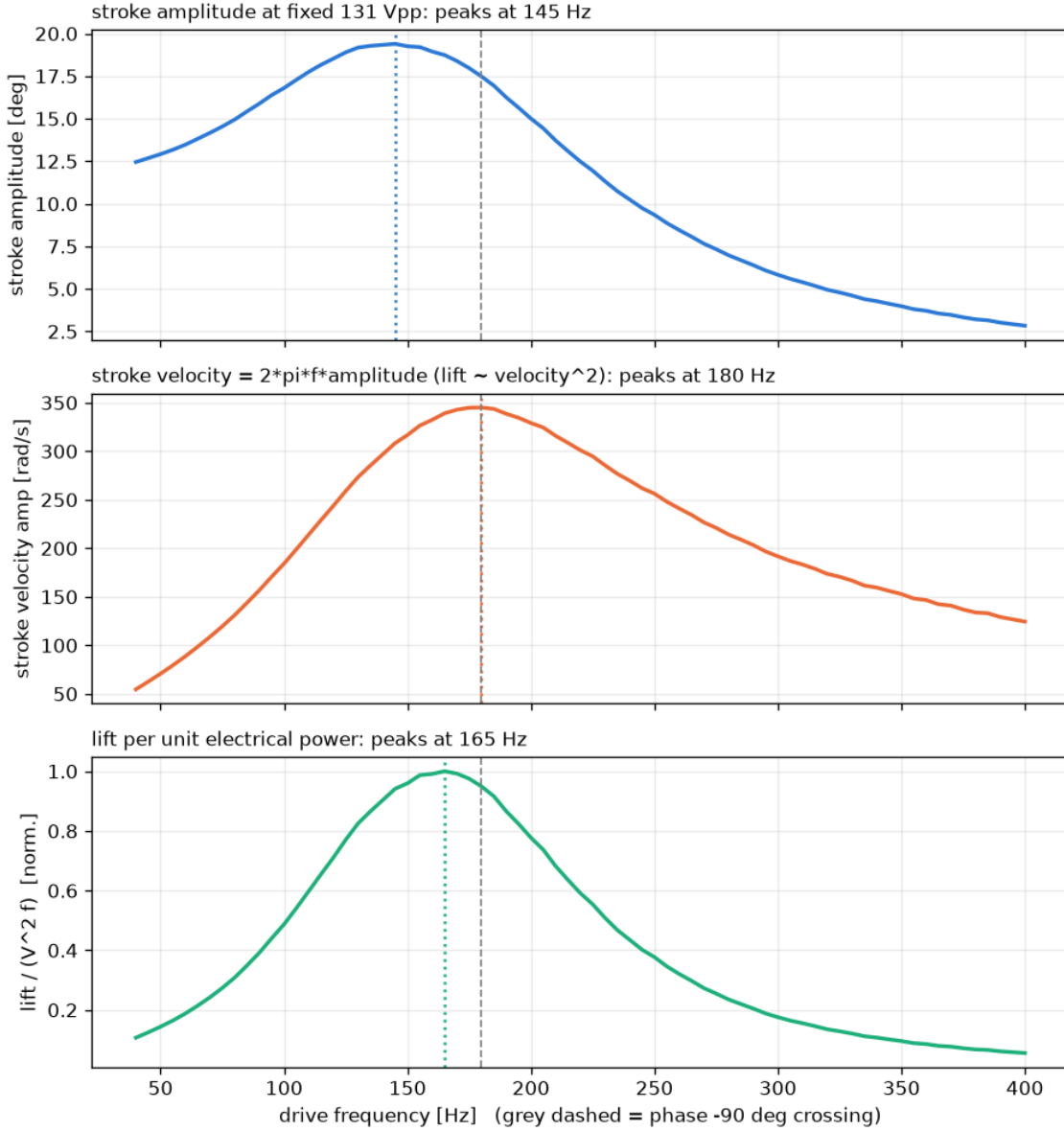


Figure 4: Frequency plots for various flight parameters, ran at 131 Vpp (rough amplitude needed for hover).

frequency is forced steady-state vibration where the system absorbs maximum energy, while natural frequency is how fast the system oscillates as its own energy dissipates.

With the geometric flexures alone the stroke-amplitude peak at flight drive came out at 100–125 Hz, well below the ~ 155 Hz resonance of the current hardware (the 181 Hz natural frequency implied by Steinmeyer’s Table I belongs to an older design and is not the target). We therefore add a lumped trim stiffness k_t^{trim} on the stroke output joint of each side only (`revolute_1.3` on the left, `revolute_1.4` on the right), not on the two coupler flexures. That joint’s angle is ϕ itself, so a spring there adds to the output stiffness one for one and the code constant is directly comparable to the paper’s lumped k_t ; a spring on a coupler joint would contribute only $\kappa (\partial \theta_i / \partial \phi)^2$, a geometric fraction that was not measured. The three geometric flexure springs per side are left unchanged.

Sizing the trim. With the geometric flexures alone the -90° phase crossing sat at 148–155 Hz and the quasi-static k_{eq} at 385 N/m. Two experiments located the deficit: doubling all six flexure springs doubled k_t^{eff} exactly ($k_{eq} 385 \rightarrow 465$ N/m, crossing $148 \rightarrow 165$ Hz), so the loop-closure constraints are not softening the linkage; doubling only the output flexure gave $k_{eq} = 433$ N/m and 158 Hz, so the output flexure carries about 60% of k_t^{eff} and the two coupler flexures 40%. The effective slider-side mass implied by the crossings is $m_{eq} \approx 0.44$ g, above the paper’s

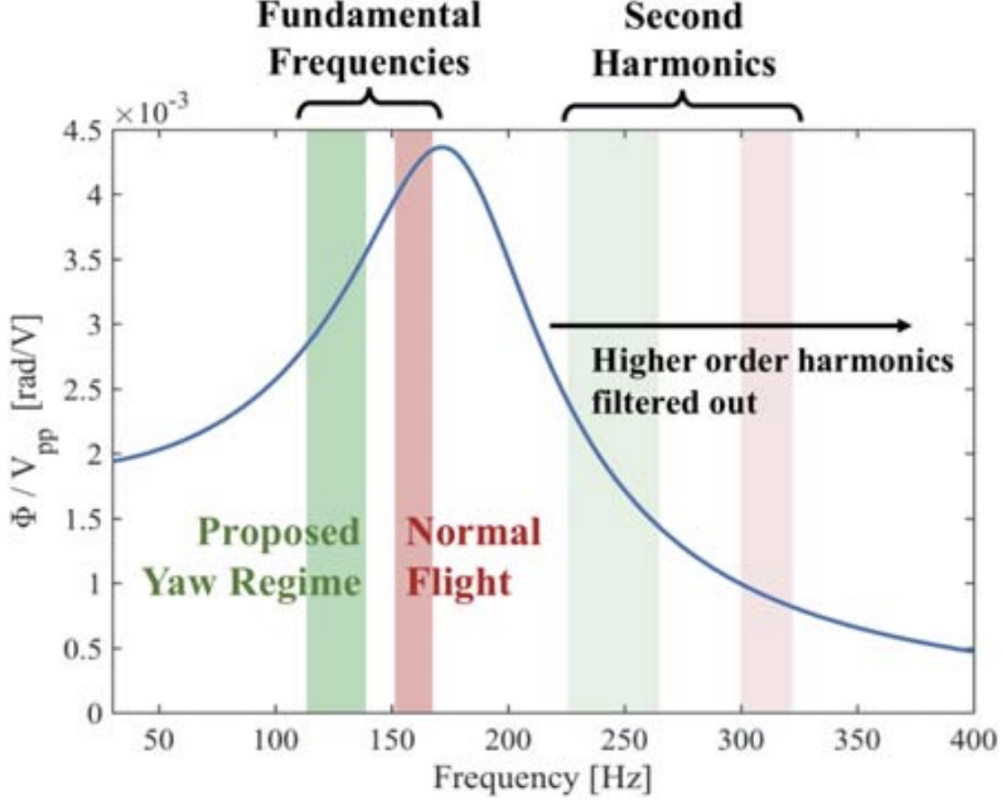


Figure 5: Freq. response plot from Steinmeyer's paper.

0.388 g because the blade-element model adds aerodynamic added mass. Wing inertia was deliberately not changed.

The target is the stroke-amplitude peak at the $131 V_{pp}$ hover drive, which on the hardware sits near 155 Hz. Because the drag damping at flight amplitude is heavy (fitted $\zeta \approx 0.4$, against 0.22 for Steinmeyer's linearized b_{eq}), the amplitude peak sits about 20% below the -90° natural frequency here, so matching a 155 Hz peak requires a natural frequency near 188 Hz. A first trim of $30 \mu\text{N m/rad}$ (2026-09-16) put the peak at 145 Hz; raising it to $40 \mu\text{N m/rad}$ (2026-09-17) put it at 155.4 Hz. The sensitivity is about $+10$ Hz of amplitude peak and $+8$ Hz of natural frequency per $10 \mu\text{N m/rad}$. If the hardware's damping is closer to Steinmeyer's, its natural frequency is only about 5% above its amplitude peak, so the two plants then differ in damping rather than stiffness; that discrepancy is left open.

Result with the $40 \mu\text{N m/rad}$ trim (Fig. 3).

- Quasi-static: $x_{ss}/\delta_{free} = 0.50$, $k_{eq} \approx 600 \text{ N/m}$, $k_t^{\text{eff}} \approx 53\text{--}63 \mu\text{N m/rad}$; low-frequency gain 0.17 deg/V ($3.0 \times 10^{-3} \text{ rad/V}$).
- Stroke-amplitude peak at 155.4 Hz for $131 V_{pp}$ (the target), 156 Hz for $100 V_{pp}$, 155 Hz for $200 V_{pp}$ and 173 Hz for $50 V_{pp}$. Peak gains 0.40 deg/V ($7.0 \times 10^{-3} \text{ rad/V}$), 0.31 (5.4×10^{-3}), 0.28 (4.9×10^{-3}) and 0.25 (4.3×10^{-3}) for 50, 100, 131 and $200 V_{pp}$.
- -90° phase crossing at 186 Hz ($100 V_{pp}$), 188 Hz ($131 V_{pp}$) and 193 Hz ($200 V_{pp}$); the $50 V_{pp}$ curve has a small kink near 180 Hz that makes its crossing ill-defined (about 182 Hz).
- At the 155 Hz, $131 V_{pp}$ hover point: stroke amplitude 18.3° , slider 0.129 mm , phase -57° . At 180 Hz, $131 V_{pp}$: 17.5° , phase -81° .
- Above the peak the roll-off is a single second-order slope with the phase reaching -175° to -180° by 400 Hz; the amplitudes converge above 280 Hz. The $50 V_{pp}$ points above 280 Hz again have up to 25% fit residual from wing-pitch harmonics at very small stroke; the other three curves stay below 2.5%.

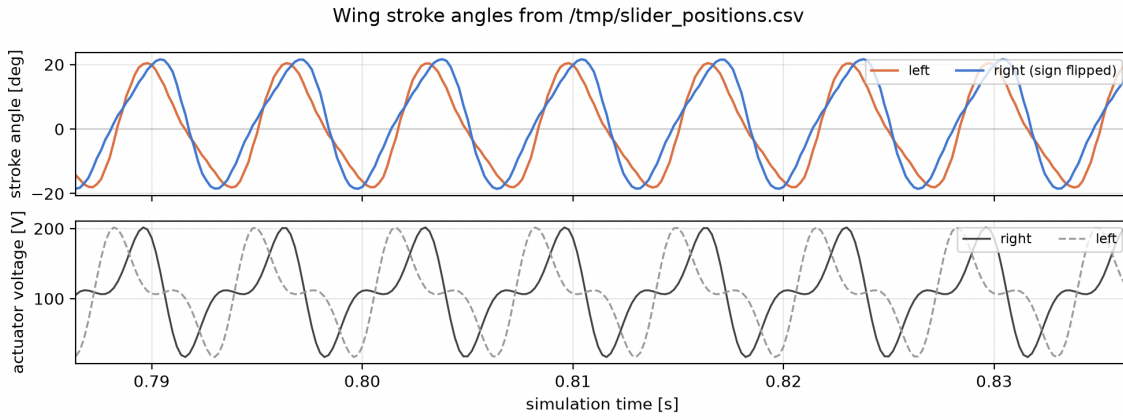


Figure 6: Input actuator voltage and resultant wing stroke angle.

Because the voltage-to-stroke gain and phase at the operating frequency changed again, the wing force maps used by the MPC must be re-identified against this plant. As of 2026-09-17 only the welded standalone binary has been rebuilt with the $40 \mu\text{N m/rad}$ trim; the flying app and both Simulink servers failed to link for lack of disk space and still carry $30 \mu\text{N m/rad}$.

2.5 Validation before sweeping

Apply a static voltage step and record the steady-state slider displacement x_{ss} and stroke angle ϕ_{ss} . Two checks follow:

$$\frac{x_{ss}}{\delta_{free}} = \frac{k_a}{k_a + T^2 k_t} = \frac{k_a}{k_{eq}} \approx 0.60, \quad (132)$$

$$\frac{\phi_{ss}}{x_{ss}} = T \approx 2666 \text{ rad/m}. \quad (133)$$

If either is far off, the spring nominal positions or the loop-closure constraint compliance are wrong and a frequency sweep would be meaningless. This test also reveals whether the SAP solver remains well behaved through the loop-closure constraints once the stiff PD gains are removed; constraint chatter at this stage is the signal to add slider damping.

Result (2026-09-16, geometric flexures only, before the output trim). Measured with the sweep tool at 20 and 40 Hz, where dynamic amplification is under 2%:

- $x_{ss}/\delta_{free} = 0.76\text{--}0.82$ (predicted 0.60), so $k_{eq} \approx 365\text{--}395 \text{ N/m}$ rather than 500, and $k_t^{\text{eff}} = (k_{eq} - k_a)/T^2 \approx 11\text{--}17 \mu\text{N m/rad}$ against the paper's lumped 28.2. The distributed flexures weighted by $(\partial\theta_i/\partial\phi)^2$ are about half as stiff at the output as the lumped value.
- $T = \phi/x \approx 2490 \text{ rad/m}$ at flight strokes (2390 at very small stroke), against 2666 in the paper.
- No constraint chatter: the sinusoid-fit residual is 1–5% of the fundamental at 100–200 V_{pp} and no slider damping was needed. The 50 V_{pp} residual rises to $\sim 18\%$ above 190 Hz and 20–50% above 280 Hz, where the stroke is only 1–6° and harmonic content from the wing-pitch hinge dominates; those small-amplitude high-frequency points are the least trustworthy in Fig. 3.

Once the sweep matches the analytic reference at small amplitude, the simulated $H(\omega)$ can be used to reproduce the split-cycle filtering result (Steinmeyer et al. Fig. 4) directly, by driving the second-harmonic waveform $V(t) = V_{amp}[a_1 \sin \omega t \pm a_2 \sin 2\omega t + a_3 \sin 3\omega t]$ and comparing the resulting $\phi(t)$ upstroke and downstroke speeds.

2.6 Order of work

1. Analytic reference curve from `get_transfer_function.m` over 50–400 Hz.

2. Plant changes (done, in both the standalone app and the Simulink TCP server as of 2026-09-16): reflected inertia, **PrismaticSpring**, pure-force slider actuation, and a **RevoluteSpring** on each of the six transmission flexures (Table 3). Before that date the Simulink server still used a PD joint servo ($K_p = 1000$, $K_d = 0.5$) tracking a prescribed slider displacement, so every closed-loop flight run over TCP prior to it used a kinematically prescribed stroke.
3. Static validation of T , k_a/k_{eq} , and the emergent k_t^{eff} (done 2026-09-16, Section 2.5).
4. Frequency flag, ϕ logging, sweep driver, Bode plot against the reference (done 2026-09-16, `tools/stroke_frequency_sweep.py`, Fig. 3).
5. Amplitude dependence (done, same figure), then the split-cycle waveform test.
6. Raise the transmission output stiffness so the simulated stroke-amplitude peak at hover drive matches the hardware's ~ 155 Hz (done 2026-09-17: $40 \mu\text{N m/rad}$ lumped trim on each stroke output joint, Section 2.4; the Simulink server and flying binaries still need rebuilding with it), then re-identify the wing force maps used by the MPC, since the Simulink server runs this plant (to do).

3 Conclusion and Open Questions

4 Appendix

4.1 Change of Basis Derivation

The derivation is a change of basis from the mixed-frame angular velocity

$$\boldsymbol{\omega} = -\dot{\phi} \mathbf{e}_{X'} + \dot{\theta} \mathbf{e}_{z'} + \dot{\psi} \mathbf{e}_x \quad (134)$$

to the wing-fixed basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$. Using the paper's sign convention,

$$\mathbf{e}_{z'} = \sin \psi \mathbf{e}_y + \cos \psi \mathbf{e}_z, \quad (135)$$

$$\mathbf{e}_{X'} = \sin \theta \mathbf{e}_x + \cos \theta (\cos \psi \mathbf{e}_y - \sin \psi \mathbf{e}_z). \quad (136)$$

Substitution gives

$$\begin{aligned} \boldsymbol{\omega} &= -\dot{\phi} [\sin \theta \mathbf{e}_x + \cos \theta (\cos \psi \mathbf{e}_y - \sin \psi \mathbf{e}_z)] \\ &\quad + \dot{\theta} (\sin \psi \mathbf{e}_y + \cos \psi \mathbf{e}_z) + \dot{\psi} \mathbf{e}_x \\ &= (\dot{\psi} - \dot{\phi} \sin \theta) \mathbf{e}_x + (-\dot{\phi} \cos \theta \cos \psi + \dot{\theta} \sin \psi) \mathbf{e}_y \\ &\quad + (\dot{\phi} \cos \theta \sin \psi + \dot{\theta} \cos \psi) \mathbf{e}_z. \end{aligned} \quad (137)$$

Therefore, the wing-fixed angular velocity components are

$$\omega_x = \dot{\psi} - \dot{\phi} \sin \theta, \quad (138)$$

$$\omega_y = -\dot{\phi} \cos \theta \cos \psi + \dot{\theta} \sin \psi, \quad (139)$$

$$\omega_z = \dot{\phi} \cos \theta \sin \psi + \dot{\theta} \cos \psi. \quad (140)$$

The $-\dot{\phi} \sin \theta$ term in ω_x appears because wing deviation projects part of the flapping angular velocity onto the wing pitch axis. In the planar case, $\theta = 0$, so $\omega_x = \dot{\psi}$.

4.2 Non-dimensionalization Derivation

Start with the dimensional spanwise integral from the lift expression:

$$I_F = \int_0^R (r + x_r)^2 c(r') dr. \quad (141)$$

Here r is the spanwise coordinate used in the blade-element integral, and r' is the same physical distance measured from the wing root for evaluating the local chord distribution. Thus, for this simplified planar wing model, we take $r' = r$.

Using the normalized dimensions for the distance along the wing root, the distance from the wing root to the rotation axis, and the chord length,

$$\hat{r} = \frac{r}{R}, \quad \hat{x}_r = \frac{x_r}{R}, \quad \hat{c}(\hat{r}) = \frac{c(r)}{\bar{c}}, \quad (142)$$

we have

$$r = R\hat{r}, \quad x_r = R\hat{x}_r, \quad c(r) = \bar{c}\hat{c}(\hat{r}), \quad dr = Rd\hat{r}. \quad (143)$$

Note that (x_r, y_r) are the coordinates of the root of the wing (which is basically the hinge). Substituting these into I_F gives

$$I_F = \int_0^1 (R\hat{r} + R\hat{x}_r)^2 \bar{c}\hat{c}(\hat{r}) R d\hat{r} \quad (144)$$

$$= R^3 \bar{c} \int_0^1 (\hat{r} + \hat{x}_r)^2 \hat{c}(\hat{r}) d\hat{r}. \quad (145)$$

Therefore, the nondimensionalized integral is

$$\hat{F} = \frac{I_F}{R^3 \bar{c}} = \int_0^1 (\hat{r} + \hat{x}_r)^2 \hat{c}(\hat{r}) d\hat{r}. \quad (146)$$

Expanding the square,

$$\hat{F} = \int_0^1 (\hat{r}^2 + 2\hat{x}_r \hat{r} + \hat{x}_r^2) \hat{c}(\hat{r}) d\hat{r} \quad (147)$$

$$= \int_0^1 \hat{r}^2 \hat{c}(\hat{r}) d\hat{r} + 2\hat{x}_r \int_0^1 \hat{r} \hat{c}(\hat{r}) d\hat{r} + \hat{x}_r^2 \int_0^1 \hat{c}(\hat{r}) d\hat{r}. \quad (148)$$

Since $\bar{c} = A/R$,

$$\int_0^1 \hat{c}(\hat{r}) d\hat{r} = \frac{1}{R\bar{c}} \int_0^R c(r) dr = \frac{A}{R\bar{c}} = 1. \quad (149)$$

Define the nondimensional radius moments by

$$\hat{r}_1 = \int_0^1 \hat{r} \hat{c}(\hat{r}) d\hat{r}, \quad (150)$$

$$\hat{r}_2^2 = \int_0^1 \hat{r}^2 \hat{c}(\hat{r}) d\hat{r}. \quad (151)$$

Then the compact form is

$$\hat{F} = \hat{r}_2^2 + 2\hat{x}_r \hat{r}_1 + \hat{x}_r^2. \quad (152)$$

5 Moment of Inertia Derivation

The moment of inertia for the left wing $I_{xx,CM} = 1.3 \times 10^{-6}$ and $I_{xy,CM} = 2.25 \times 10^{-7}$, from the CAD model, and around the center of mass. All other elements of the matrix can be approximated as 0 for this pitch equation. However, these need to be expressed around the pitch hinge axis. Assuming the CAD axes are already aligned with the wing-fixed axes used in the derivation, the tensor parallel axis theorem tells us that

$$I_H = I_{CM} + m ((\mathbf{d}^T \mathbf{d}) \mathbf{1} - \mathbf{d} \mathbf{d}^T), \quad (153)$$

where m is the wing mass, $\mathbf{1}$ is the 3×3 identity matrix, and $\mathbf{d}$ is the displacement vector from the wing center of mass to the pitch hinge origin, expressed in the same wing-fixed frame as I_{CM} . The displacement vector is

$$\mathbf{d} = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} = \begin{bmatrix} -7.6171 \\ -0.5949 \\ 0.000 \end{bmatrix}. \quad (154)$$

All distances must be in units consistent with the CAD inertia units. If the inertia is used in SI units, then the displacement vector should be converted from millimeters to meters before applying the theorem.

Expanding the parallel-axis term gives

$$(\mathbf{d}^T \mathbf{d}) \mathbf{1} - \mathbf{d} \mathbf{d}^T = \begin{bmatrix} d_y^2 + d_z^2 & -d_x d_y & -d_x d_z \\ -d_y d_x & d_x^2 + d_z^2 & -d_y d_z \\ -d_z d_x & -d_z d_y & d_x^2 + d_y^2 \end{bmatrix}. \quad (155)$$

Therefore, the hinge-frame inertia components needed by the pitch equation are

$$I_{xx,H} = I_{xx,CM} + m(d_y^2 + d_z^2), \quad (156)$$

$$I_{xy,H} = I_{xy,CM} - m d_x d_y. \quad (157)$$

Substituting the displacement vector above,

$$I_{xx,H} = 1.3 \times 10^{-6} + m((-0.5949)^2 + 0^2), \quad (158)$$

$$I_{xy,H} = 2.25 \times 10^{-7} - m(-7.6171)(-0.5949). \quad (159)$$

Equivalently, using the millimeter CAD coordinates as written,

$$I_{xx,H} = 1.3 \times 10^{-6} + 0.35390601 m, \quad (160)$$

$$I_{xy,H} = 2.25 \times 10^{-7} - 4.53141279 m. \quad (161)$$

These are the constants to use for I_{xx} and I_{xy} in the pitch dynamics equation:

$$I_{xx} = I_{xx,H}, \quad (162)$$

$$I_{xy} = I_{xy,H}. \quad (163)$$

If the CAD inertia tensor is reported in a frame that is rotated relative to the wing-fixed frame, first rotate it into the wing-fixed frame,

$$I_{CM}^{wing} = R I_{CM}^{CAD} R^T, \quad (164)$$

and then apply the parallel axis theorem above.